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Development and optimization of human glucocerebrosidase-encoding mRNA for Gaucher disease therapy

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ABSTRACT

Gaucher disease (GD) is a rare autosomal-recessive lysosomal storage disorder caused by mutations in the *GBA1* gene encoding the lysosomal hydrolase glucocerebrosidase (GCase). Mutations in GCase lead to glucosylceramide accumulation within macrophages. Current treatments, including enzyme replacement therapy (ERT) and substrate reduction therapy (SRT), alleviate symptoms but are limited by high cost, frequent dosing, and adverse effects, highlighting the need for novel strategies. To achieve higher expression levels and improved stability, we designed and optimized a series of *hGBA1*-mRNA by changing untranslated regions (UTRs), codon usage, and poly(A) tails, and evaluated their performance in vitro and in vivo. Optimized constructs achieved >6-fold higher GCase activity compared with the least efficient variants 24 h post-transfection in HEK293T and RAW264.7 cells, with an average half-life exceeding 54 h. The expressed enzyme localized to lysosomes and restored normal morphology and substrate accumulation in *GBA1*-knockout (KO) HEK293T cells. Following a single administration of *hGBA1*-mRNA encapsulated in lipid nanoparticles (LNPs) in wild-type FVB mice, GCase activity was detectable in the liver and spleen within 72 h. Our results demonstrate that optimized *hGBA1*-mRNA-LNPs can deliver functional human GCase in vivo, providing a promising and efficient mRNA-based therapeutic approach for GD.

1. Introduction

Gaucher disease (GD), one of the most common lysosomal storage disorders, is a rare autosomal recessive disorder caused by mutations in the *GBA1* gene [1]. It was first described by Philippe Gaucher in a patient with an enlarged spleen in 1882 [2,3]. The *GBA1* gene, located on chromosome 1 (1q22), comprises 12 exons and encodes a 497-amino-acid β-glucocerebrosidase (GCase) precursor. Naive GCase undergoes protein processing and glycosylation in the endoplasmic reticulum and Golgi complex and is then trafficked to lysosomes as a mature enzyme [4]. In lysosomes, GCase hydrolyses glucosylceramide (GlcCer) to glucose and ceramide, functioning in glycosphingolipid metabolism [5]. More than 700 *GBA1* mutations have been recorded by the NCBI database, and these mutations cause a significant decrease in GCase activity [6], resulting in the accumulation of GlcCer and glucosylsphingosine (GlcSph) in macrophages, which are called Gaucher cells [1,5]. GD is categorized into three types, with type 1 (GD1) classically defined as the non-neuronopathic form. However, patients with GD1 carry a significantly increased risk of developing Parkinson's disease later in life, highlighting a critical neurological implication of GCase deficiency beyond the classical classification [7].

There are two main clinical treatments for GD: enzyme replacement therapy (ERT) and substrate reduction therapy (SRT) [5,8]. ERT drugs, the mainstream drugs for GD patients [9], are designed to supply functional GCase through intravenous (i.v.) injection of recombinant GCase. Three ERT products, Cerezyme, Velaglucerase alfa, and

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Taliglucerase alfa, were approved by the Food and Drug Administration in 1994, 2010, and 2012, respectively [10]. ERT is recognized as the "gold" treatment for GD1 and GD3 patients [9], however, it has several disadvantages [11]. Firstly, the high cost of ERT drugs, e.g., for a US patient, the annual cost is $200,000 [12]. Secondly, the half-life of recombinant proteins is short and requires frequent, lifelong administration– once every two weeks, and each infusion lasts for 1-2 h [12,13], which may cause psychological pressure for patients. Thirdly, ERT may induce anti-drug antibodies [14], potentially reducing the therapy effects and, in some cases, causing hypersensitivity reactions, including anaphylaxis [15]. The most important challenge is that GCase cannot cross the blood-brain barrier (BBB), limiting the benefit for patients with neurological impairments [14]. Therefore, novel therapies for GD are necessary, and mRNA therapy may represent a promising option for GD patients [16].

mRNA typically contains a 5′ untranslated region (UTR), an open reading frame (ORF), a 3' UTR, and a poly(A) tail, encoding the specific protein, and is being developed as a therapeutic strategy to treat a variety of genetic diseases [17]. In contrast to exogenously manufactured recombinant proteins (e.g., ERT), therapeutic proteins made from administered mRNA may better mimic their endogenous synthetic pathways [18], since mRNA can be directly expressed in the host cells with the same post-translational modifications. Moreover, the encoded signal peptide can direct nascent proteins into the correct intracellular trafficking route and organelle, whereas lysosomal delivery of exogenous enzyme from ERT can be inefficient. Remarkably, mRNA treatment produces transient protein levels while reducing the risks of genomic integration and off-target effects compared with DNA-based gene therapy or gene editing [18]. In the current study, we developed an mRNA-LNP approach to supply functional human GCase in vivo. We comprehensively optimized the mRNA constructs with various element sequences to improve GCase expression and activity. The function of *hGBA1*-mRNA was demonstrated in a GBA1 knockout (*GBA1*-KO) cell model and in wild-type FVB mice.

2. Materials and methods

2.1. HEK293T GBA1-KO assay

To knock out the *GBA1* gene in HEK293T, we used the three-plasmid combination system as previously published [19], comprising a target selector, a frame selector, and a universal donor plasmid. The target selector expresses a specific sgRNA to cleave the *GBA1* gene, whereas the frame selector and the universal donor plasmid carry a puromycin resistance gene and a terminator (Supplementary Fig. 1). Briefly, to generate a cell line with *GBA1*-KO, HEK293T cells were co-transfected with the *GBA1* sgRNA plasmid, the universal donor plasmid, and the frame selector sgRNA at a plasmid of 1:2:1. 48 h after transfection, 3 μg/mL puromycin was added, and selection of *GBA1* KO cells was continued for an additional 7 days. Individual colonies were picked, and single clones were isolated by limited dilution. Homozygous gene-edited clones were identified by genomic PCR and immunoblotting using antibodies against *GBA1* and GAPDH.

2.2. Animals

Wild-type FVB male mice (4-6 weeks old) and male C57BL/6J mice (6 weeks old) were purchased from GemPharmatech (China). Sex was not expected to materially affect mRNA-LNP uptake following i.v. administration; therefore, only male mice were used. All animal experiments complied with relevant ethical regulations and were ethically approved by the Animal Experiment Center of Southern University of Science and Technology. Mice were housed under a 12 h light/dark cycle with ad libitum access to standard chow and water. The mice received a single injection of *hGBA1*-mRNA-LNP (1.5 mg/kg) through the tail vein or retro-orbital sinus and were euthanized at 6, 12, 24, 36, and 72 h post-injection. Mice of the control group were given the same volume of PBS (n = 3). Liver and spleen were harvested for protein expression and enzymatic activity detection, while serum was collected for anti-GCase antibody detection. Protein concentrations were determined using a BCA Protein Assay kit (Thermo Fisher Scientific, Waltham, MA, USA; CAT# 23225).

2.3. GCase plasmid construction

The GCase expression plasmid, containing a CMV enhancer, CMV promoter, T7 promoter, 5′ UTR, the native human GCase coding sequence with a C-terminal FLAG tag, a 3' UTR, and a poly(A) tail, was synthesized by Genewiz. Plasmids were amplified in *E.coli* DH5α and purified by FastPure EndoFree Plasmid Maxi Kit (Vazyme Biotech, Nanjing, China). Plasmid concentrations were detected by Nanodrop spectrophotometry. For fluorescence-based visualization, GCase was fused to RFP or mCherry at the N-terminus.

2.4. Linear DNA synthesis

Linear DNA was generated by cleaving the plasmid with the 2 μl Xhol enzyme at 37 °C for 4 h, followed by purification and ethanol precipitation using 3 M sodium acetate and anhydrous ethanol. DNA concentrations were measured by NanoDrop spectrophotometry. The purified linear DNA was used as the template for in vitro transcription.

2.5. In vitro mRNA synthesis and encapsulation

mRNA encoding human wild-type GCase was synthesized in vitro from a linear DNA template using T7 RNA polymerase-mediated transcription. During this process, uridine was completely substituted by 5-methoxyuridine. The mRNA was purified using RNA Cleanup kit (New England Biolabs, Beverly, MA, USA, CAT# T2050L), and its concentration and integrity were determined by Nanodrop spectrophotometry and agarose gel electrophoresis, respectively.

Lipid nanoparticles (LNPs) were prepared by an ethanol-drop nanoprecipitation method as described previously. Briefly, lipids dissolved in ethanol at a molar ratio of 50:10:38.5:1.5 (ionizable:helper: structural: PEG) were mixed with acidified mRNA at a 3:1 lipid:mRNA ratio via a microfluidic device. Formulations were dialyzed against phosphate-buffered saline (PBS; pH 7.4) for 4 h using dialysis cassettes, concentrated using Amicon ultra centrifugal filters (EMD Millipore), and stored at 4 °C until use. All formulations were characterized for particle size and mRNA encapsulation efficiency, which were found to be between 100 ± 20 nm in size, with >90 % encapsulation.

2.6. Tissue distribution analysis

To evaluate the expression of GCase in vivo, mCherry was fused to the hGBA1 plasmid. Then, mCherry-hGBA1-mRNA was synthesized and encapsulated in LNP as mentioned above. Mice received a single i.v. tail-vein injection of 1.5 mg/kg mCherry-hGBA1-mRNA-LNP and were euthanized at 6 h and 12 h post-injection. Excised liver tissues were imaged using an in vivo imaging system (IVIS) Spectrum imaging system.

2.7. Cell transfection

One day prior to transfection, cells were seeded into 6 or 12-well plates at 5×10^5 and 1×10^5 cells per well, respectively, resulting in ~80 % confluency on the day of transfection. Cells were transfected with 1 μg of mRNA or plasmids using Lipofectamine 2000 or Lipofectamine 3000 (Invitrogen) by following the manufacturer's protocol, whereas mRNA encapsulated with LNP was added in culture media. 24-48 h post-transfection, cells were harvested and used for protein expression or enzymatic activity measurement. Total protein concentrations were

determined by the BCA Protein Assay kit (Thermo Fisher Scientific, Waltham, MA, USA, CAT# 23225).

For the KO cells assay, 1 μg of Q1, 1 μg of sgRNA, and 2 μg of puro-donor plasmids were co-transfected. 24-48 h post-transfection, cells were cultured in DMEM containing 10 % FBS, 1 % P/S, and 3 μg/mL puromycin. Survived cells were harvested and passaged through for subsequent assays.

2.8. qRT-PCR

After transfection with 0.4 μg/mL of different *GBA1* mRNA formulations in 12-well plates, cells were harvested at 6 h, 12 h, 32 h, and 54 h post-transfection. Total RNA was extracted using an RNA extraction kit (Vazyme, Nanjing, China, CAT# RC112) according to the manufacturer's instructions. Five hundred nanograms of total RNA was used to produce cDNA using a reverse transcription kit (Cwbio, Jiangsu, China, CAT# CW3371 M).

Intracellular GCase mRNA levels were quantified by the standard SYBR Green-based qRT-PCR (Cwbio, Jiangsu, China, CAT# CW3360S) using primers for GCase, GCase-O1, GCase-O2, and GAPDH (primer sequences listed in the Supplementary Table 1). For the reaction, 40 ng of the cDNA was subjected to amplification for 35 cycles on a qTOWER3 system (Analytik Jena, Germany), and threshold cycle (Ct) values were processed and analyzed using the qPCRsoft Software. Relative mRNA abundance was calculated using the 2^(-ΔΔCt) method, with GAPDH as the endogenous control. The ΔCt value obtained from GCase mRNA was normalized against the Ct value obtained from GAPDH mRNA. Fold change is reported relative to the control group.

2.9. Western blot

All cell lysates and liver tissues were used for the Western blot. Cells were lysed in RIPA buffer (Beyond, Shanghai, China) supplemented with a protease inhibitor cocktail (Beyond, Shanghai, China) on ice for 5 min, followed by centrifugation to clarify lysates. Tissues (~50 mg) were homogenized in 1% Triton-X100 supplemented with 1% protease inhibitor cocktail, sonicated (4 °C, 30 s × 6), and clarified by centrifugation (12,000×*g*; 10 min, 4 °C). The protein concentration was measured using the BCA Protein Assay kit (Thermo Fisher Scientific, Waltham, MA, USA, CAT# 23225) according to the manufacturer's instructions. Equal amounts of protein (~20 μg) were separated by 12% SDS-PAGE and transferred onto methanol-activated PVDF membranes (Bio-Rad, Hercules, CA, USA) using the semi-dry transfer method facilitated by Trans-Blot Turbo Transfer System (Bio-Rad, Hercules, CA, USA). Membranes were blocked in 5% (w/v) skim milk in 1 × phosphate-buffered saline with Tween 20 (PBST) for 2 h at room temperature and then incubated with primary antibodies for 1 h. The membrane underwent five washes using 1 × PBST before the incubation with the appropriate HRP-conjugated secondary antibody for 1 h at room temperature. Bands were developed using ECL western blotting reagent and imaged using a ChemiDoc Imaging System (SHT, Shanghai, China). After detection of target proteins, membranes were stripped using stripping buffer (1 % Tween 20, 0.1 % SDS, pH 2.2 in distilled water), washed twice, and subsequently washed three times with 1 × PBST containing 5 % skim milk. Subsequently, the membranes were stained for GAPDH using the procedures described above. Mouse Anti-DDDDK tag antibody (Bioss Antibodies, Beijing, China; CAT# bsm-33346 M) was used at a 1:10,000 dilution. The HRP-conjugated GAPDH monoclonal antibody (Proteintech, Wuhan, China; CAT# HRP-60004) was used at a 1:10,000 dilution. Human GBA Antibody was used in a 1:2000 dilution (Thermo Fisher Scientific, Waltham, MA, USA; CAT# PA5-67940). An HRP-conjugated goat anti-rabbit antibody (Abcam, Cambridge, MA, USA; CAT# ab6721) was applied at a 1:2000 dilution.

2.10. GCase activity assay

All cell lysates and liver tissues were used for the GCase activity assay. Liver tissues were treated as mentioned above. For the GCase activity assay, the artificial fluorogenic substrate 4-methylumbelliferyl-beta-D-glucopyranoside (APExBIO) was used as described previously [20]. Briefly, the assay mixtures were pre-incubated with or without 1 mM conduritol-B-epoxide (CBE) for 30 min at room temperature (InvivoChem, USA; CAT# 6090-95-5), and the substrate 4-Methylumbelliferyl β-D-glucopyranoside (4MUG) was added and incubated for 1 h at 37 °C. The reaction was stopped by adding 1 M glycine. 100 μl of the assay mixture was measured using a standard fluorometric assay in a flat-bottom 96-well plate. (Ex = 355 nm, Em = 460 nm, 0.1 s). The activity was calculated by subtracting the fluorescence of the CBE-treated control from that of the active reaction and then normalizing to the total protein amount. Results are presented as RFU/min/μg protein.

2.11. Immunocytochemistry

To inspect the shapes of lysosomes in *GBA1*-KO HEK293T cells and the colocalization of GCase with lysosomes, cells were subjected to immunostaining with various antibodies, tailored to the specific objectives of each experiment. HEK293T cells, which were grown on glass coverslips in 24-well plates or on glass-bottom 24-well plates (P24-1.5H–N, Cellvis), underwent fixation with 4% paraformaldehyde (PFA) for 10 min. They were then permeabilized using a buffer containing 0.3 % Igepal CA-630, 0.05 % Triton-X 100, and 0.1 % IgG-free BSA in PBS for 5 min. Subsequently, the cells were blocked with a buffer comprising 0.05 % Igepal CA-630, 0.05 % Triton-X 100, and 5 % normal goat serum in PBS for 60 min. The cells were incubated with primary antibodies, anti-Flag (66008-4-Ig, Proteintech) and anti-GCase (PA5-27436, Invitrogen), at room temperature for 1 h. Subsequently, secondary antibodies, Goat anti-Rabbit Alexa Fluor 568 (A-11011, Invitrogen) and Goat anti-Mouse Alexa Fluor 568 (A-11004, Invitrogen), were applied for 1 h at room temperature after washing three times for 5 min each with PBS. The nucleus was stained with DAPI. Cells were washed three times with wash buffer (0.05 % Igepal CA-630, 0.05 % Triton-X 100, 0.2 % IgG-free BSA in PBS) and twice with PBS. Coverslips were mounted using ProLong Gold Antifade Reagent. Dip the coverslip in ddH2O before mounting to prevent salt contamination. Images were acquired with a Zeiss LSM980 confocal microscope using a 63 × Apochromat oil-immersion objective and analyzed using ImageJ.

2.12. Statistical analysis

All numerical data are presented as mean ± standard deviation (SD). The sample size (n) represents the number of biologically independent animals or cell lysates. Normality of the GCase activity data for each tissue was confirmed using the Shapiro-Wilk test (all $p > 0.05$). Comparisons between the treatment and control groups for each sample were performed using unpaired, two-tailed Student's t-tests and One-way ANOVA test. Multiple comparisons were performed using Tukey's multiple comparisons test. All analyses were conducted using GraphPad Prism 9.5.0. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

3. Results

3.1. Identification of mRNA constructs encoding human GCase and subcellular localization in HEK293T

To evaluate the effect of mRNA sequences encoding human GCase, we first designed two *GBA1* DNA constructs, out of which one is a natural human *GBA1* gene, named h*GBA1* (Fig. 1a), and another is a human natural *GBA1* gene lacking the N-terminal 39 amino acids, named short-h*GBA1* (Fig. 1b). We transcribed *GBA1* DNAs in vitro and detected

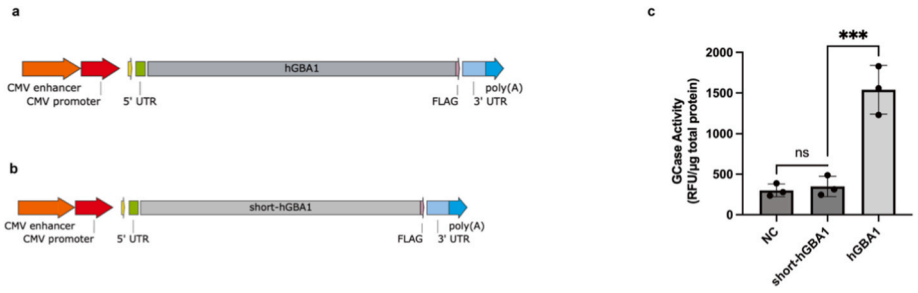


Fig. 1. DNA structure of *GBA1* and expression activity. The DNA sequence encodes the complete amino acid sequence of natural human GCase (a) and residues 40-495 of natural human GCase (b). Relative GCase activity in HEK293T cells at 24 h after transfection with *hGBA1*-mRNA and short-*hGBA1*-mRNA (c). n = 3; data are presented as mean ± SD (ns, no significant; ***, $p < 0.001$, one-way ANOVA).

GCase activity in the mRNA-transfected HEK293T cells. However, only the *hGBA1*-mRNA can express a functional GCase while short-*hGBA1*-mRNA showed activity comparable to the negative control (Fig. 1c). To achieve higher expression level and stability of hGBA1-mRNA, we designed variant sequences through codon optimization (without altering the amino acid sequence), and by screening different 5′ UTRs, 3′ UTRs, and poly(A) tails based on the length, secondary structure, and reported efficacy of these elements in enhancing mRNA stability and translational efficiency. Specifically, for the poly(A) tail, we incorporated designs informed by recent studies, such as the work by Yi Kuang and colleagues, which suggests that incorporating nucleotide spacers (e.g., C) within the poly(A) sequence can further improve mRNA expression and stability [21,22] (Supplementary Table 2). The differences of *hGBA1*-mRNA expressed GCase activity in HEK293T are given in Fig. 2a.

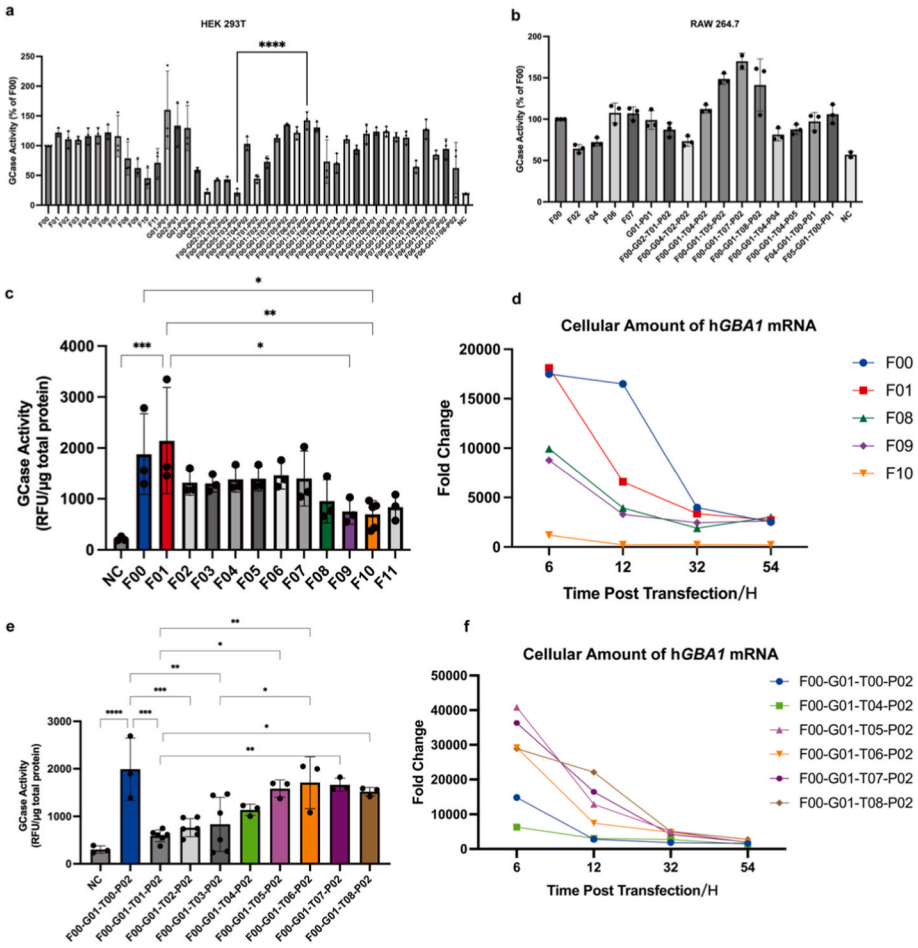


Fig. 2. The effects of codon optimizations, substitutions of 5′ UTR, 3′ UTR, and poly(A) tail on *hGBA1*-mRNA expression. (a) Relative GCase activity for HEK293T cells at 24 h post-transfection with different *GBA1* mRNAs. Values were normalized to the activity of F00, which was set at 1. n = 3; data are presented as mean ± SD. (b) Relative GCase activity for RAW 264.7 cells at 24 h post-transfection with different *GBA1* mRNAs. Values were normalized to the activity of F00, which was set at 1. n = 3; data are presented as mean ± SD. (c) Relative GCase activity for HEK293T cells at 24 h post-transfection with substitution of 5′ UTR. n = 3; data are presented as mean ± SD. (d) Cellular amount of *hGBA1*-mRNA with substitution of 5′ UTR for HEK293T cells at 6 h, 12 h, 32 h, and 54 h post-transfection. (e) Relative GCase activity for HEK293T cells at 24 h post-transfection with substitution of 3′ UTR. n = 3; data are presented as mean ± SD. (f) Cellular amount of *hGBA1*-mRNA with substitution of 3′ UTR for HEK293T cells at 6 h, 12 h, 32 h, and 54 h post-transfection. (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$; one-way ANOVA.).

F00-G01-T07-P02 showed an improvement in activity by > 6-fold compared to F00-G05-T03-P02 in HEK293T cells. To further explore the impact of different *hGBA1*-mRNA in different cells, we also evaluated the activity of the top and lowest eight out of the total mRNAs in RAW 264.7 cells (Fig. 2b), which had the same trend compared with HEK293T cells. Next, we detected the cellular amount of *hGBA1*-mRNA at different times post-transfection in HEK293T (Fig. 2d and f). Substitutions of different 5′ UTR influenced the GCase activity by affecting the stability of *hGBA1*-mRNA in HEK293T, while substitutions of different 3' UTR did not influence the GCase activity by affecting the stability of *hGBA1*-mRNA in HEK293T (Fig. 2c–f). Moreover, the decay curves were fitted to calculate the mRNA half-life ($t_{1/2}$), respectively (Supplementary Table 3). This table clearly demonstrates that F00-T07-P02 offers the most favorable combination of enhanced mRNA stability and high functional enzyme output, which forms the core quantitative rationale for its selection as our lead candidate. Also, we calculated the folding energy of *hGBA1*-mRNAs according to the website (http://rna.tbi.univie.ac.at), with no significant differences (Table 1). The quality of *hGBA1*-mRNA-LNP is given in Table 2.

Here, we selected F00-G01-T07-P02 to evaluate the subcellular localization of exogenous *hGBA1*-mRNA in wild-type and *GBA1*-KO HEK293T cells. Co-localization of hGCase protein with LAMP-1 (a lysosome marker) confirmed the appropriate lysosomal localization of GCase encoded by *hGBA1*-mRNA (Fig. 3c). The effect of *hGBA1*-mRNA was analyzed in HEK293T cells via knocking out the *GBA1* gene. As shown in Fig. 3e, knocking out the *GBA1* gene caused an increasing volume of lysosomes known as lysosomal storage (Fig. 3e, green). The cellular GBA1 mRNA level and the activity of GCase are significantly decreased (Fig. 3f and g). Next, we transfected equally 1 μg/mL *hGBA1*-plasmid and *hGBA1*-mRNA in wild-type and HEK293T KO *GBA1* cells. This resulted in significantly higher expression and enzymatic activity of GCase compared to cells that were not transfected, as evidenced by the recovery in lysosome volume (Fig. 3e). Next, we assessed the functions of *hGBA1*-mRNA encapsulated in LNP with the components of SM102, cholesterol, DSPC, and PEG-DMG in HEK293T cells. Fig. 3a and Table 2 show the good encapsulation efficiency and size distributions of *hGBA1*-mRNA-LNP. As shown in Fig. 3d, transfection with *hGBA1*-mRNA and h*GBA1*-mRNA-LNP resulted in overall high expression and enzymatic activity both in HEK293T cells, while *hGBA1*-mRNA-LNP had higher expression and enzymatic activity than *hGBA1*-mRNA (Fig. 3b and d). These data suggest that the *hGBA1*-mRNA expresses robust levels of GCase protein within the suitable subcellular compartment in cells.

3.2. Efficacy of hGBA1-mRNA-LNP in vivo

To verify the expression of *hGBA1*-mRNA-LNP in vivo, we designed a subclone of *hGBA1*-mRNA encoding GCase fused with mCherry peptide for mouse tissue imaging. Wild-type C57BL/6 mice were i.v. administered with 1.5 mg/kg of mCherry-*hGBA1*-mRNA-LNP and sacrificed at 6 h and 12 h. Successful GCase protein expression was detected in the liver (Fig. 4d and e). Notably, signals were not uneven (Fig. 4d at 6 h), and it might be the result of injection leakage during administration. We next performed i.v. administration in wild-type FVB mice with 1.5 mg/kg of *hGBA1*-mRNA-LNP and sacrificed at 6 h, 12 h, 24 h, 36 h, and 72 h, respectively, and harvested livers, spleens, and serum. The control group was administered the same volume of PBS as *hGBA1*-mRNA-LNP (n = 3). Liver and spleen tissues were ultrasonicated, and the supernatants were used for protein and enzyme activity analysis. GCase expression and activity were all observed in livers (Fig. 4b and c) and spleens (Supplementary Fig. 2) post-injection at 6 h, 12 h, 24 h, 36 h, and 72 h.

Table 2
Quality of hGBA1-mRNA-LNP.

Encapsulation efficiency	Particle size (d. nm)	pdI
95.85 %	97.39 ± 10	0.138 ± 0.4

To evaluate whether i.v. administration of *hGBA1*-mRNA generates neutralizing anti-GCase antibodies, we measured anti-GCase antibody levels (Supplementary Fig. 3) and did not detect a notable difference between mice injected with *hGBA1* and PBS, suggesting no significant innate immune activation associated with acute inflammation was observed at the early 24-48 h time point post-administration, consistent with early-phase expression before adaptive immunity onset.

4. Discussion

This study describes the design, optimization, and initial preclinical validation of an mRNA-LNP therapeutic candidate for GD. We show that our engineered hGBA1-mRNA construct drives robust expression of enzymatically active human-GCase both in vitro and in vivo after systemic administration in wild-type mice, supporting further development of this platform.

Our work aligns with and extends recent progress in the field. A pioneering study by Liu et al. validated the mRNA-LNP platform in GD using a murine GBA1 sequence, demonstrating prolonged enzymatic activity compared to ERT in a genetic mouse model (GBA1 D427V) [23], highlighting a key pharmacological advantage over protein-based therapies. Our study advances this concept in two ways. First, by employing the human *GBA1* coding sequence, we directly enhance clinical translatability. Second, through systematic optimization of UTRs, codon usage, and poly(A) tail, we improved mRNA translation and stability, yielding a > 6-fold increase in protein output in vitro. While other high-yield production strategies for recombinant GCase, such as lentiviral stable cell lines utilizing computer-based algorithms, codon optimization, and strong promoters with different *GBA1* variants [24–26], adeno-associated virus-mediated gene therapy [27] have been successfully developed, the mRNA-LNP platform offers distinct advantages for expression pattern and therapeutic delivery. Unlike viral or stable cell-based systems, mRNA-LNP delivery provides transient but highly tunable expression, avoids genomic integration risks, and enables rapid, scalable manufacturing [28–30]. Notably, our data indicate that the expressed GCase remained intracellular, aligning with the required lysosomal localization for function, a critical mechanistic feature supporting its therapeutic strategy.

Despite these strengths, we acknowledge several crucial limitations that outline essential future work. A key limitation of this preclinical study is the use of wild-type mice for initial in vivo efficacy assessment. The use of wild-type mice, while ideal for initial proof-of-concept, does not recapitulate the GD pathological environment. Efficacy must be rigorously validated in established GD animal models to assess true disease-modifying potential. Another important consideration is the preliminary nature of our immunogenicity assessment. Our data demonstrate the absence of an acute humoral response following a single administration in immunocompetent mice. However, this short-term, single-dose study cannot rule out the potential for adaptive immune responses or antibody formation upon chronic, repeated dosing. Therefore, a comprehensive long-term study with multiple administrations is an essential next step to fully characterize the immunogenicity risk profile before clinical translation. Meanwhile, the therapeutic

Table 1
Folding kinetics of different mRNAs.

mRNA	F00	F08	F09	F10	G01-P02	F00-G01-T05-P02	F00-G01-T06-P02	F00-G01-T07-P02	F00-G01-T08-P02
Free energy	−726.5	−717.7	−720.4	−720	−668.5	−643.8	−661	−646.4	−666.8

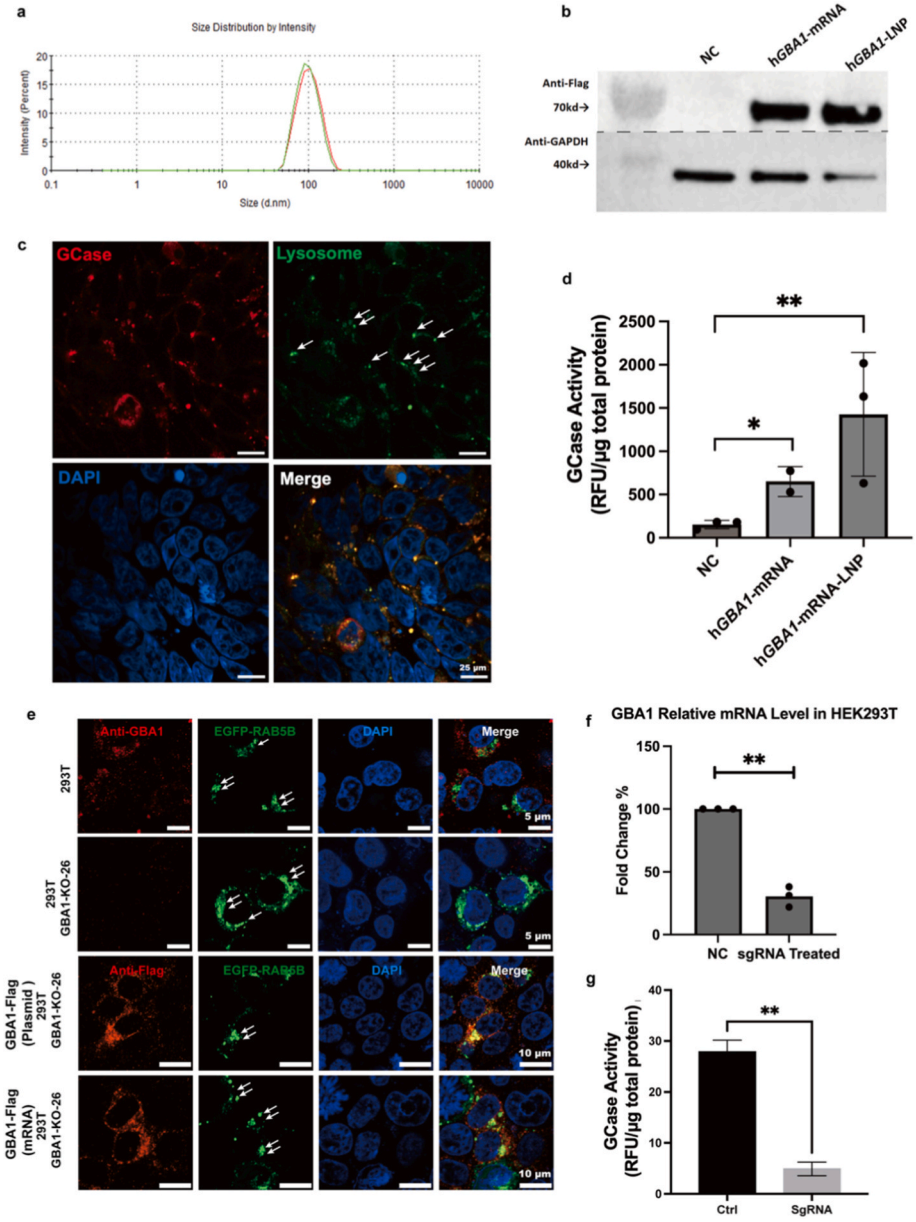


Fig. 3. The effect of *hGBA1*-mRNA in wild-type and *GBA1*-KO HEK293T cells and the quality of *hGBA1*-mRNA-LNP. (a) Size distribution by intensity of *hGBA1*-mRNA-LNP. (b) Western blot showed the amount of GCase after 24 h transfection of *hGBA1*-mRNA and *hGBA1*-mRNA-LNP. The blots for anti-flag (upper panel) and for anti-GAPDH (lower panel) were from different membranes with the same amount of sample (Dashed separators denote non-sequential lanes, and the whole uncropped blots are provided in the supplementary materials). (c) Confocal immunofluorescence analysis with 24 h post-transfection of *hGBA1*-mRNA with RFP fusion in HEK293T cells with a lysosome tracker (Lyso Tracker). White arrows represent lysosomes. (d) Relative GCase activity for HEK293T cells at 24 h post-transfection of *hGBA1*-mRNA and *hGBA1*-mRNA-LNP. (e) Confocal immunofluorescence analysis with 24 h post-transfection of *hGBA1*-mRNA in *GBA1*-KO HEK293T cells. Wild-type, *GBA1*-KO HEK293T cells, *GBA1*-KO HEK293T transfected with *hGBA1*-mRNA, were both processed for three-label immunofluorescence for anti-*GBA1*, anti-flag, EGFP-RAB5B (an endo-lysosome marker), and DAPI. Line 1: wild-type HEK293T; line 2: *GBA1*-KO HEK293T; line 3: *GBA1*-KO HEK293T were transfected with *GBA1* plasmid (1 μg/mL); line 4: *GBA1*-KO HEK293T were transfected with *hGBA1*-mRNA (1 μg/mL). Representative confocal images of individual cells are shown with the merged images in the right column. White arrows represent lysosomes. (f) Relative cellular GBA1 mRNA level in *GBA1*-KO HEK293T cells. (g)Relative GCase activity for *GBA1*-KO HEK293T cells. (*, $p < 0.05$; **, $p < 0.01$; Student's t-tests).

efficacy of such mRNA platforms may be constrained by the limited translational capacity and resource competition within target cells, potentially leading to suboptimal or unsustainable protein expression. Intriguingly, recent advances in cellular engineering provide a complementary approach to overcome this bottleneck. Pisani et al. computationally identified small molecules, such as Filgotinib, that rewire cellular resource allocation and significantly boost expression of various exogenous genetic payloads, including those delivered by mRNA, without additional genetic modifications [31]. This highlights the critical concept that optimizing the host cell's production environment is as important as delivering the therapeutic blueprint.

Looking forward, the therapeutic potential of this platform can be further augmented through integrated optimization of both delivery and expression. A paramount direction is the engineering of next-generation LNPs for cell-specific targeting, such as to macrophages [32], and for blood-brain barrier [33–35] penetration to address all GD subtypes. This targeted approach would enhance drug bioavailability at disease sites while minimizing off-target effects. Beyond delivery precision, enhancing intracellular protein yield is equally critical. Combining such computational-based algorithms with optimized, targeted mRNA-LNP

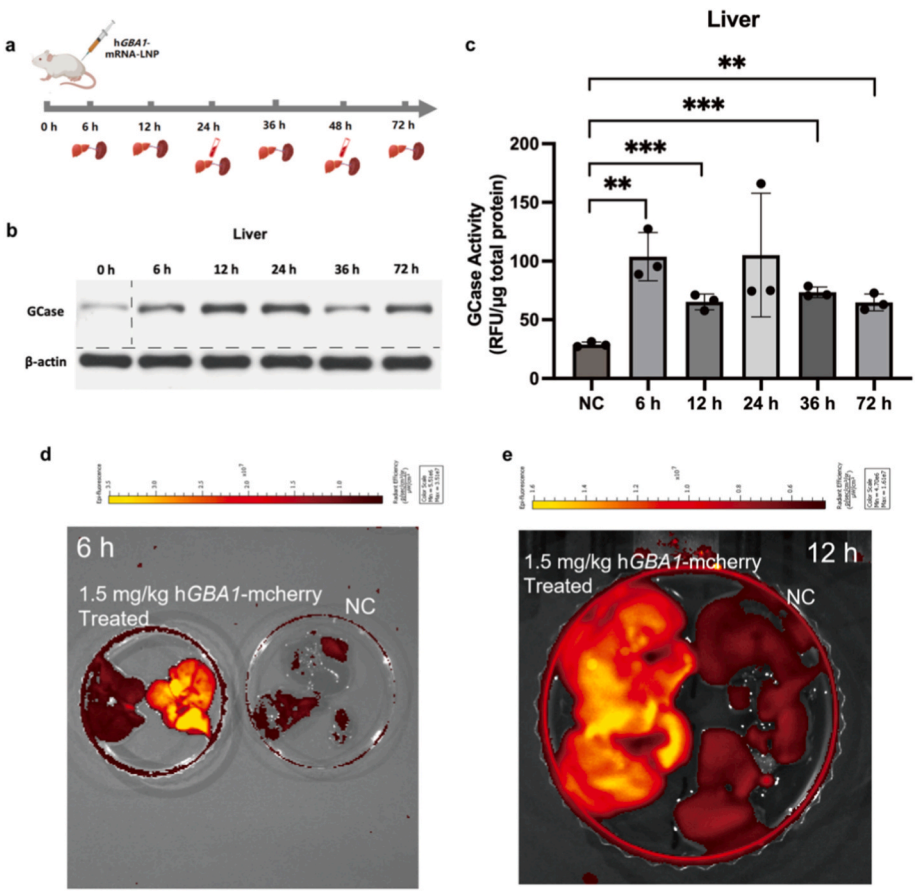


Fig. 4. The effect of *hGBA1*-mRNA-LNP in wild-type mice. (a) A scheme of the animal experiment. (b) Western blot showed the expression of GCase in liver tissues post-injection of 1.5 mg/kg *hGBA1*-mRNA-LNP in wild-type FVB mice at 0 h, 6 h, 12 h, 24 h, 36 h, and 72 h. The blots for GCase (upper panel) and for β-actin (lower panel) were from different membranes with the same amount of sample. Among them, lane 1 (0 h) and lane 2 (6 h) were not contiguous in the blot (Dashed separators denote non-sequential lanes, and the whole uncropped blots are provided in the supplementary material). (c) Relative GCase activity in liver tissues post-injection of 1.5 mg/kg *hGBA1*-mRNA-LNP in wild-type FVB mice at 0 h, 6 h, 12 h, 24 h, 36 h, and 72 h n = 3; data are presented as mean ± SD. (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; Student's t-tests). (d) The IVIS image of liver tissues post 6 h and 12 h (e) vein injection of 1.5 mg/kg mCherry-*hGBA1*-mRNA-LNP in wild-type C57BL/6 (4-6 weeks).

formulation represents a compelling, multi-pronged strategy. This synergistic approach could improve dosing efficiency and ultimately achieve a superior therapeutic outcome for patients with GD.

5. Conclusions

This study focuses on applying GCase-encoding mRNA for treating GD as a potential alternative therapy to overcome the limitations of current treatments, such as ERT and SRT. mRNA approach as a protein replacement therapy may represent the transformative power of biomedical innovation, potentially setting a new standard for genetic disorder treatments. The implications of mRNA technology extend beyond the GD community, paving the way for developing similar therapies for a spectrum of genetic diseases.

Statement

All procedures were performed in compliance with relevant laws and institutional guidelines and have been approved by Southern University of Science and Technology on April 21, 2022, and the acceptance number is SUSTech-SL2022041001.

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CRediT authorship contribution statement

Shunping Feng: Conceptualization, Data curation, Investigation, Visualization, Writing – original draft. **Xiaoming Jiang:** Methodology. **Nino Rcheulishvili:** Visualization, Writing – original draft, Writing – review & editing. **Dimitri Papukashvili:** Visualization, Writing – original draft, Writing – review & editing. **Shuaiyang Jing:** Methodology. **Pan Zhu:** Methodology. **Xiaxia Song:** Methodology. **Xiaohui Li:** Methodology. **Yang Ji:** Conceptualization, Supervision. **Peng George Wang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2026.153594.

Data availability

All data supporting the findings of this study are available in the paper and its Supplementary Information section. GCase, GCase-O1, GCase-O2, and GAPDH primer sequences are provided in Supplementary Table 1. Modified 3′ UTR, 5’ UTR and ORF sequences are provided in Supplementary Table 2. Plasmid sequences used for HEK293T KO GBA1 are provided in Supplementary Table 4.

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